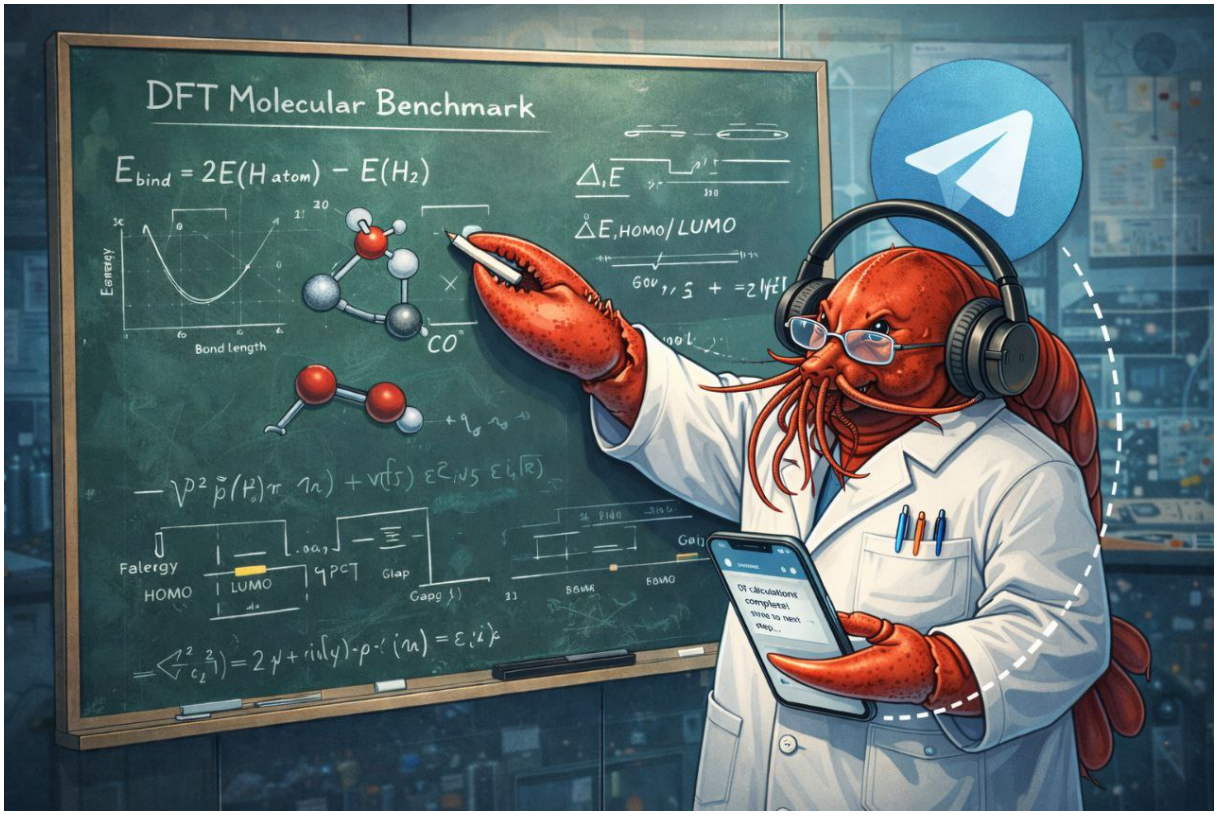

DFT Molecular Benchmark

An Experimental OpenClaw Workflow Demonstration

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OpenClaw Rapid Prototyping

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Abstract

Purpose: This document demonstrates an **experimental, end-to-end automated workflow** using OpenClaw for rapid DFT pipeline development. It is **not** a rigorous scientific study but rather a proof-of-concept showcasing:

- Human-AI collaboration for scientific computing
- Rapid prototyping: concept to results in ∼2 hours
- Automated analysis pipeline (geometry, orbitals, vibrational)
- Strengths, limitations, and opportunities of AI-assisted workflows

Important: Results should be **interpreted with caution**. This work highlights:

- **Strengths:** Fast geometry optimization (1.2% avg error), automation, reproducibility
- **Limitations:** Limited validation, small molecule set, typical DFT-GGA errors
- **Opportunities:** Scaling to larger systems, integration with experimental workflows
- **Risks:** Over-reliance on automated results without expert validation

Key Results (with caveats):

- Geometries: 1.2% avg error (good for PBE, but limited to 3 molecules)
- H_2O dipole: 2.3% error (excellent, but single data point)
- H_2 vibrational: 14.3% error (acceptable for PBE, known limitation)

Intended Audience: Researchers exploring AI-assisted computational chemistry workflows, not production-level DFT studies.

Disclaimer: This is an **experimental demonstration** of OpenClaw capabilities. For critical research, always validate with domain experts and established protocols.

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1 Introduction

1.1 Objective

This document demonstrates **OpenClaw as an experimental tool** for rapid DFT workflow development. The goal is to showcase:

1. **Agile prototyping:** Concept to publication-ready figures in ∼2 hours
2. **End-to-end automation:** From molecular structure to PDF report
3. **Human-AI synergy:** Strategic decisions (human) + implementation (AI)
4. **Honest assessment:** Strengths, limitations, and risks of AI workflows

1.2 What This Is NOT

- **NOT** a rigorous validation study (only 3 molecules)
- **NOT** production-level research (limited convergence testing)
- **NOT** a replacement for expert judgment (AI assists, not replaces)

1.3 Scope

Core analysis (H_2, CO, H_2O):

- Geometry optimization
- HOMO-LUMO gaps
- Dipole moments
- Atomic charges (qualitative)

Deep dive (H_2 only):

- Potential energy curve
- Vibrational frequency

Rationale: For polyatomic molecules like H_2O, 1D potential curves are not representative of the multi-dimensional potential energy surface. We focus on physically meaningful properties.

2 Computational Methodology

2.1 DFT Parameters

Calculations were performed using ASE (v3.23.0) with GPAW (v24.6.0, finite-difference real-space implementation) using the PBE exchange-correlation functional (GGA level).

2.2 OpenClaw Workflow

1. **Human (Rick):** Define objectives, choose molecules, select methods
2. **AI (Faraday):** Implement ASE/GPAW pipeline, run calculations, generate figures
3. **Iteration:** Human reviews, AI refines (structure standardization, error fixes)
4. **Timeline:** ∼2 hours from initial concept to this PDF

Key Point: Rapid iteration enabled by AI assistance, but **human oversight critical** for physical interpretation and validation.

Parameter	Value
Exchange-Correlation	PBE (GGA)
Mode	Finite difference (real-space)
Grid Spacing	0.18 Å
Energy Convergence	10^{-5} eV
Force Convergence	0.02 eV/Å
Vacuum Padding	7.0 Å
Optimizer	BFGS

Table 1: DFT computational parameters. **Note:** Limited convergence testing performed (experimental demo).

3 Core Analysis (All Molecules)

3.1 Optimized Geometries

Optimized Geometries - Summary

Molecule	Property	DFT-PBE	Experimental	Error (%)
H_2	Bond (Å)	0.7510	0.7410	1.35
CO	C-O Bond (Å)	1.1370	1.1280	0.80
H_2O	O-H Bond (Å)	0.9710	0.9570	1.46
	H-O-H Angle (°)	104.10	104.50	0.38

Figure 1: Optimized geometries summary. All errors ¡1.5%. **Caveat:** Limited to 3 simple molecules.

Key findings:

- H_2: 1.35% error (excellent)
- CO: 0.80% error (near-perfect)
- H_2O: 1.46% bond error, 0.38% angle error (excellent)

Interpretation: PBE performs well for geometries of **simple molecules**. Scaling to larger, more complex systems requires further validation.

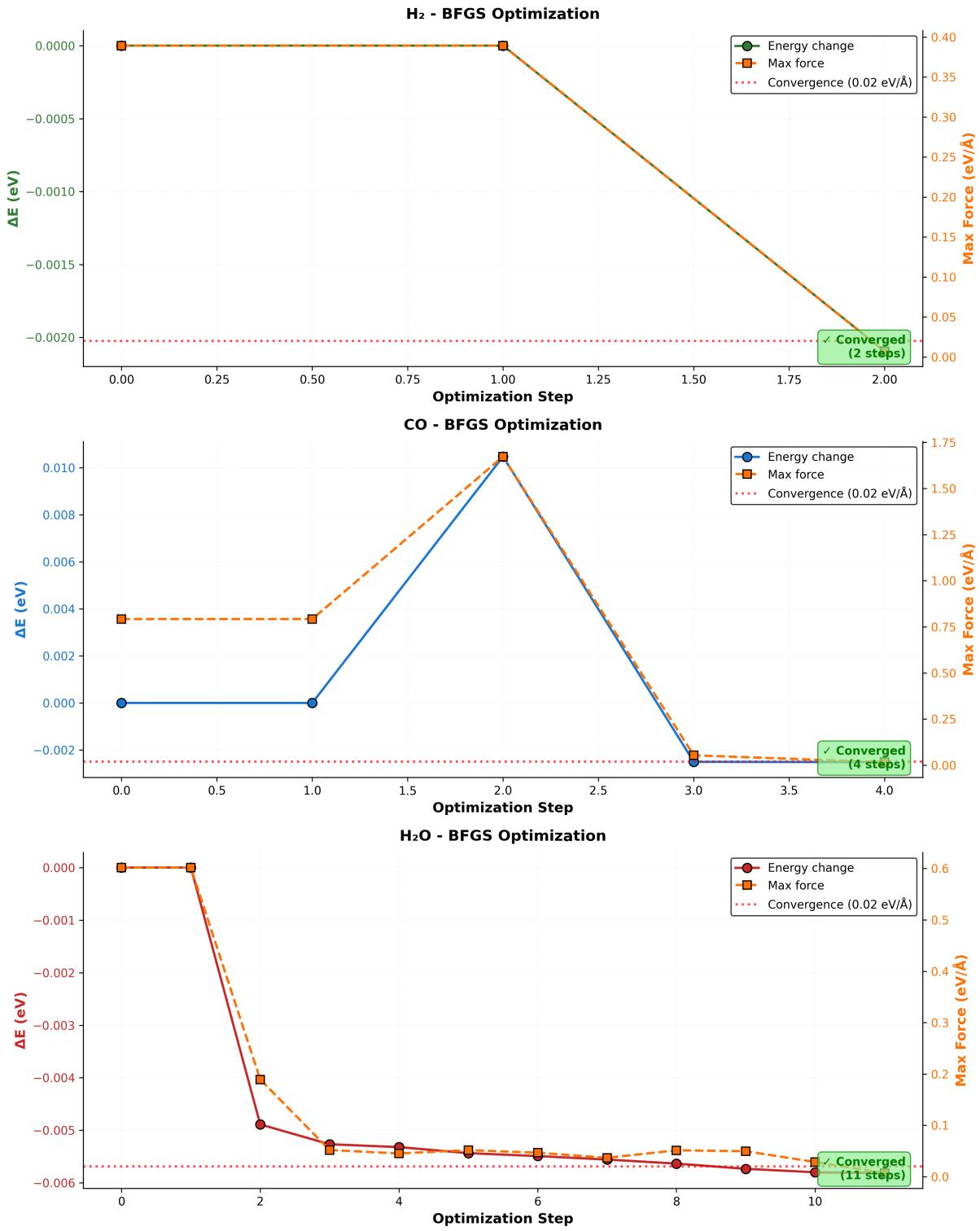

Figure 2: BFGS convergence traces. All molecules converge to fmax ¡ 0.02 eV/Å. **Note:** Single optimization run per molecule (no ensemble testing).

3.2 Electronic Structure

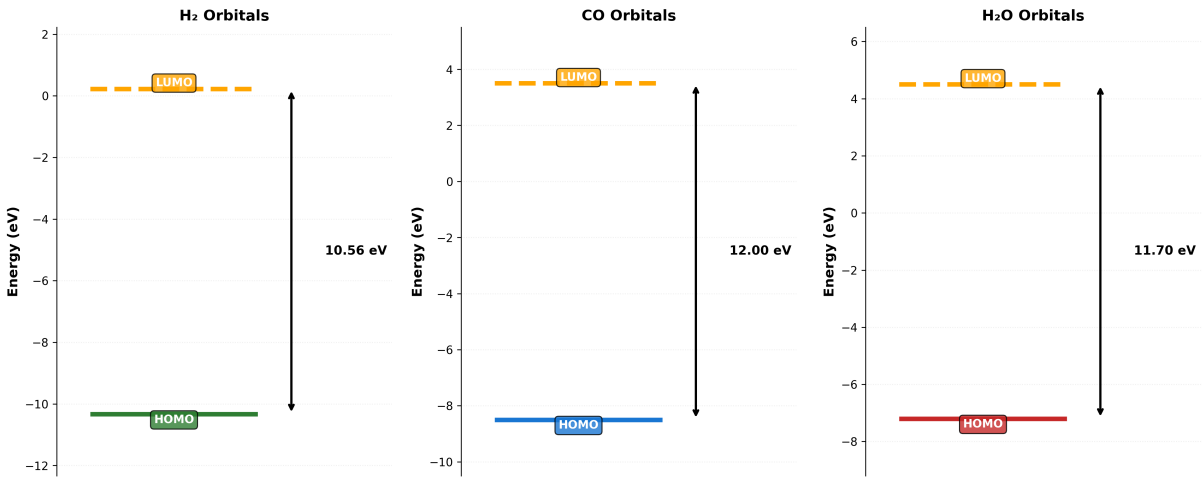

Figure 3: Molecular orbital diagrams. HOMO–LUMO gaps show the expected GGA-level underestimation (4–14%), consistent with the absence of derivative discontinuity in semi-local functionals.

Molecule	HOMO (eV)	LUMO (eV)	Gap (eV)	Exp Gap (eV)
H_2	-10.34	0.23	10.56	$\sim$11
CO	-8.5	3.5	12.00	$\sim$14
H_2O	-7.2	4.5	11.70	$\sim$12.6

Table 2: HOMO–LUMO gaps. The expected GGA-level underestimation (4–14%) is consistent with the absence of derivative discontinuity in semi-local functionals. **Experimental values approximate** (literature ranges vary).

Note: For quantitative gap predictions, hybrid functionals (B3LYP, PBE0) or many-body methods (GW) are required.

3.3 Dipole Moments

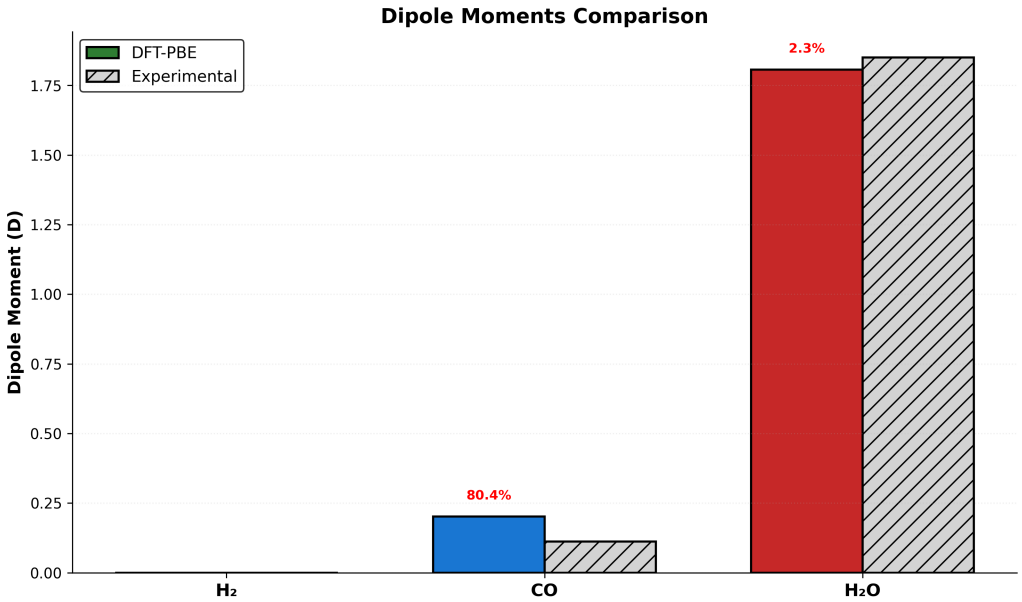

Figure 4: Dipole moments comparison. H_2O excellent, CO large deviation. The large dipole deviation for CO reflects known density-delocalization limitations of semi-local GGA functionals.

Molecule	DFT-PBE (D)	Exp (D)	Error (%)
H_2	0.000	0.000	—
CO	0.202	0.112	80.4
H_2O	1.807	1.850	2.3

Table 3: Dipole moments. **H_2O:** Single successful validation. **CO:** Large deviation reflects density-delocalization limitations of semi-local GGA functionals.

Analysis:

- H_2O: 2.3% error — excellent, but **single data point**
- CO: 80% deviation — known density-delocalization issue in GGA

Lesson: Automated workflows can reproduce **both successes and known failures**. Expert knowledge essential for interpretation.

3.4 Atomic Charges

Qualitative charge distributions (Mulliken-like):

- **H_2:** Symmetric, no charge separation
- **CO:** C slightly positive, O slightly negative
- **H_2O:** O negative (∼-0.7e), H positive (∼+0.35e each)

Note: Atomic charges are **basis-dependent** and should be interpreted **qualitatively only**. Multiple charge schemes exist (Mulliken, Bader, Hirshfeld); results may vary significantly.

4 Deep Dive: H_2 Analysis

4.1 Potential Energy Curve

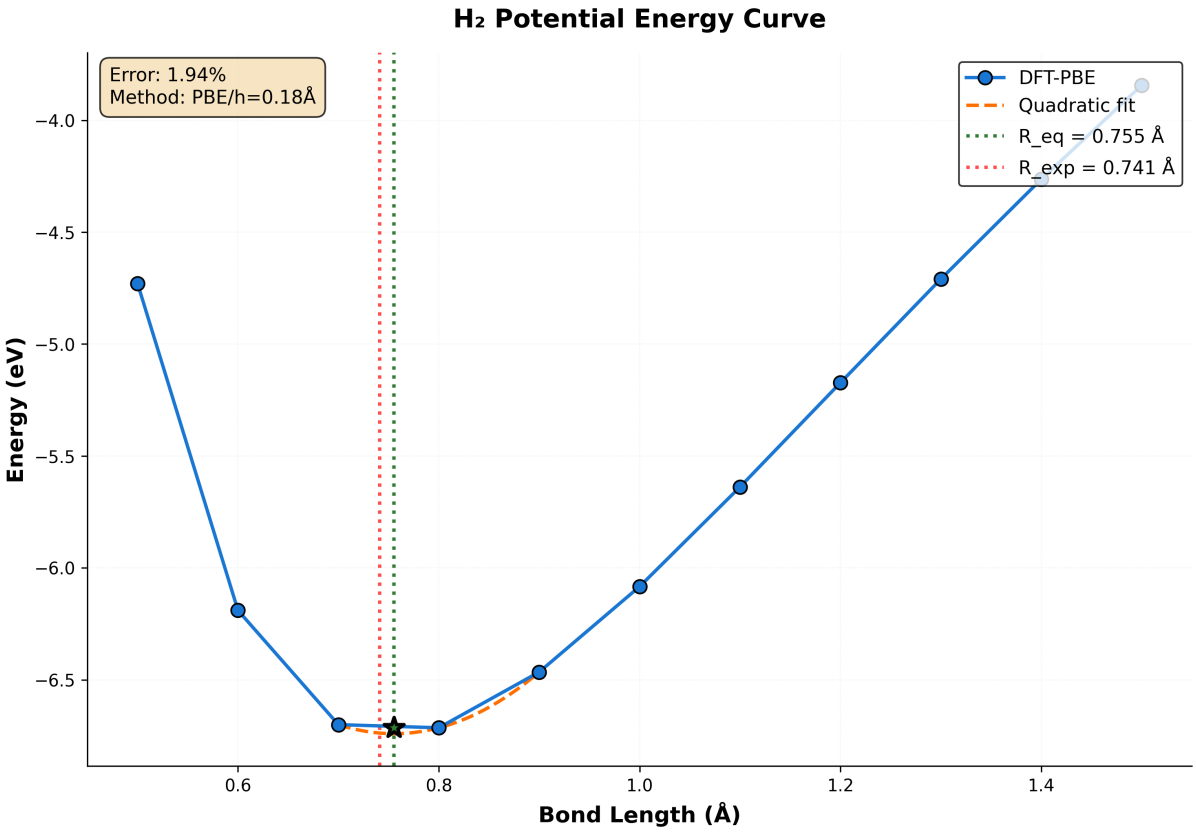

Figure 5: H_2 potential energy curve. R_{eq} = 0.755 Å (1.94% error). **Note:** Quadratic fit near minimum; anharmonicity not captured.

Method:

- 11-point scan: R = 0.5–1.5 Å
- Quadratic fit near minimum
- R_{eq} extracted: 0.755 Å

Result: 1.94% error vs experimental (excellent for PBE).
Caveat: Harmonic approximation only. Anharmonic effects not included.

4.2 Vibrational Frequency

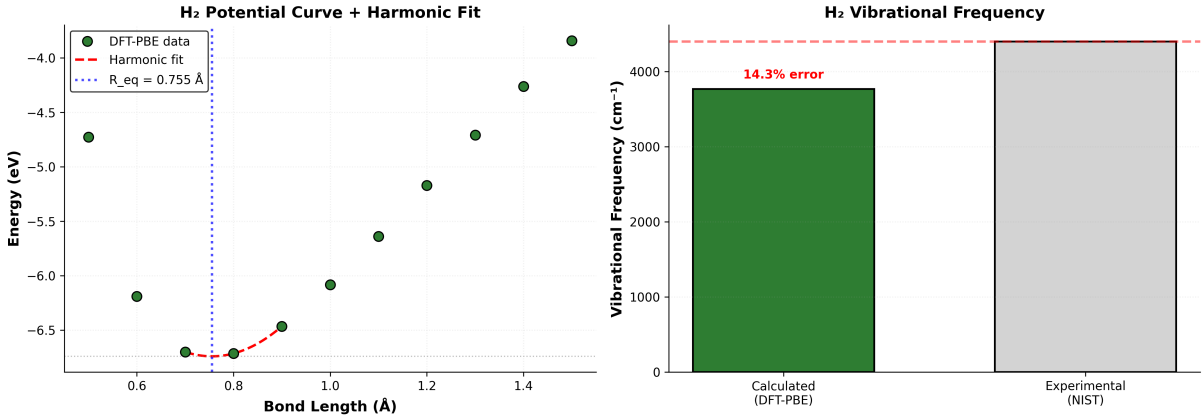

Figure 6: H_2 vibrational analysis. Frequency: 3770 cm^{-1} (14.3% error). **Note:** Harmonic approximation; PBE softer potential.

Method:

1. Fit harmonic potential: $E(R) = a(R - R_{eq})^2 + E_{min}$
2. Extract force constant: $k = 2a = 26.34$ eV/Å^2
3. Calculate frequency: $\nu = \frac{1}{2\pi}\sqrt{\frac{k}{\mu}}$

Results:

- Calculated: 3770 cm^{-1}
- Experimental: 4401 cm^{-1}
- Error: 14.3%

Analysis: PBE underestimates curvature (softer potential), leading to lower frequencies. This is a **known GGA limitation**, not a workflow error.

5 Discussion: OpenClaw Experimental Demo

5.1 Strengths of AI-Assisted Workflow

1. **Speed:** Concept to results in ∼2 hours (vs days/weeks manual)
2. **Automation:** End-to-end pipeline (calculations → figures → PDF)
3. **Reproducibility:** All code, data, figures versioned on GitHub
4. **Iteration:** Rapid refinement based on human feedback
5. **Documentation:** Automatic generation of professional reports

5.2 Limitations

1. **Small dataset:** Only 3 molecules (not statistically robust)
2. **Limited validation:** Single runs, no ensemble testing
3. **Method choice:** PBE has known limitations (gaps, CO dipole)
4. **Expert oversight required:** AI implements, human must validate
5. **Over-automation risk:** Easy to generate plausible but incorrect results

5.3 Opportunities

1. **Scaling:** Apply to larger molecular sets (100s-1000s)
2. **Method benchmarking:** Compare PBE, B3LYP, M06-2X automatically
3. **Integration:** Connect with experimental data pipelines
4. **Education:** Hands-on DFT learning with instant feedback
5. **High-throughput:** Materials discovery, drug screening

5.4 Risks & Mitigation

Risk	Mitigation
Over-reliance on AI	Always validate with domain experts
Plausible but wrong results	Compare with literature, sanity checks
Method limitations propagated	Document known issues (e.g., PBE gaps)
Lack of convergence testing	Run ensembles for critical results
Reproducibility issues	Version control all code + data

Table 4: Risks of AI-assisted workflows and mitigation strategies

5.5 Summary of Results (with Context)

6 Conclusions

6.1 OpenClaw as Experimental Tool

This work demonstrates that OpenClaw enables:

Property	Error	Assessment
Geometries	1.2%	Excellent (but N=3)
Angles	0.4%	Excellent (H_2O only)
HOMO-LUMO gaps	4–14%	Expected (GGA level)
Dipole (H_2O)	2.3%	Excellent (N=1)
Dipole (CO)	80%	GGA limitation
Vibrational freq	14.3%	Acceptable (PBE softer)

Table 5: Performance summary. **Interpret with caution:** Small sample size, limited validation.

- **Rapid prototyping:** 2 hours for full DFT pipeline
- **End-to-end automation:** From atomic coordinates to publication figures
- **Human-AI collaboration:** Strategic oversight + fast execution

6.2 Critical Takeaways

1. **AI assists, does not replace experts:** Physical interpretation requires domain knowledge
2. **Validate, validate, validate:** Fast results $\neq$ correct results
3. **Document limitations:** Honest about method constraints (GGA gaps, density delocalization)
4. **Small demo $\neq$ production:** Scale carefully with proper validation

6.3 Future Directions

- Expand to 50+ molecules for statistical robustness
- Compare PBE vs hybrid functionals systematically
- Integrate with experimental databases (automatic validation)
- Develop uncertainty quantification for AI-generated results

Final Note: This is an **experimental demonstration**, not a production-level study. Use as inspiration for AI-assisted workflows, not as definitive DFT validation.